

Sparse-to-Complete: From Sparse Image Captures to Complete 3D Scenes

YIYANG SHEN, State Key Lab of CAD&CG, Zhejiang University, China

YIN YANG, University of Utah, USA

KUN ZHOU, State Key Lab of CAD&CG, Zhejiang University, China and Hangzhou Research Institute of Holographic and AI Technology, China

TIANJIA SHAO*, State Key Lab of CAD&CG, Zhejiang University, China and Hangzhou Research Institute of Holographic and AI Technology, China

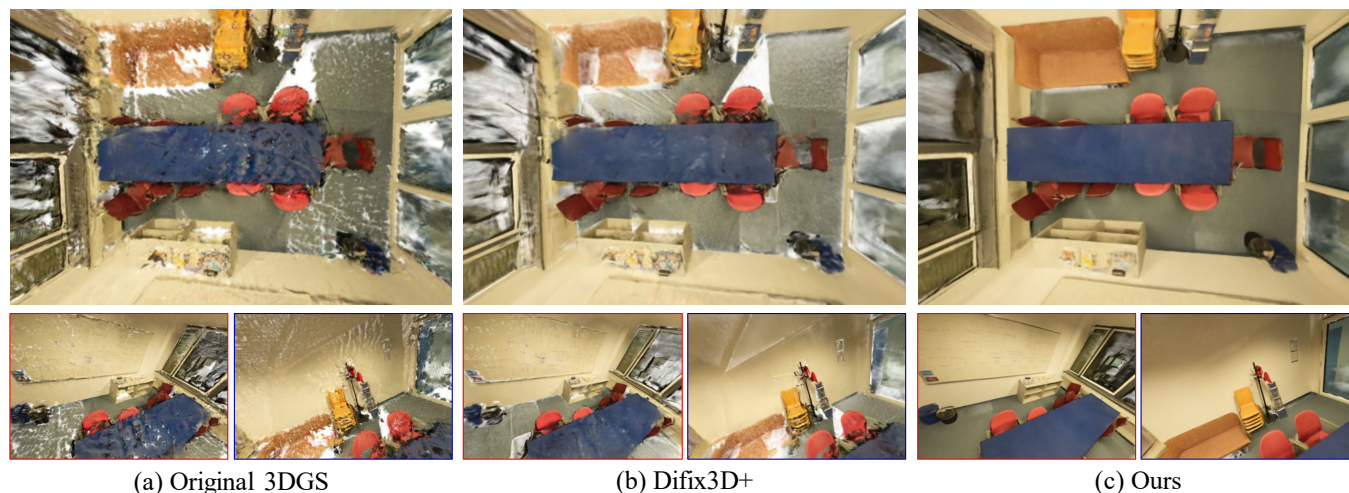


Fig. 1. A room reconstructed by the original 3DGS [Kerbl et al. 2023], Difix3D+ [Wu et al. 2025b], and our method, using only 8 images. Compared to the state-of-the-art method Difix3D+, our approach effectively recovers high-quality scenes without missing regions, blurring, or severe artifacts.

We introduce S2C-3D, a novel sparse-view 3D reconstruction framework for high-fidelity and complete scene reconstruction from as few as six to eight images. Our framework features three components: a specialized diffusion model for scene-specific image restoration, a training-free view-consistency conditioned sampling process in the diffusion model for refined Gaussian optimization, and a camera trajectory planning scheme to ensure comprehensive scene coverage. The specialized diffusion model is developed by finetuning a pretrained architecture on the input views and their corresponding degraded counterparts. The adaptation to the scene distribution allows the model to repair Gaussian renderings while effectively eliminating domain gaps. Meanwhile, the trajectory planning scheme optimizes scene coverage

by connecting each newly sampled camera to its two nearest neighbors. By iteratively constructing paths and retaining only those that significantly enhance visibility, the scheme establishes a trajectory that covers the entire scene. To address multi-view conflicts, the view-consistency conditioned sampling process quantifies the consistency between neighboring repaired images. This information is injected as a condition into the sampling process of the frozen diffusion model, facilitating the generation of view-consistent images without additional training. Consequently, our approach produces high-fidelity 3D Gaussians that are robust to artifacts. Experimental results demonstrate that S2C-3D outperforms state-of-the-art methods, constructing high-quality scenes that are free from missing regions, blurring, or other artifacts with very sparse inputs. The source code and data are available at <https://gapszju.github.io/S2C-3D>.

*Corresponding author

Authors' Contact Information: Yiyang Shen, shenyiyang114@gmail.com, State Key Lab of CAD&CG, Zhejiang University, Hangzhou, China; Yin Yang, yangzzzy@gmail.com, University of Utah, Salt Lake City, USA; Kun Zhou, kunzhou@acm.org, State Key Lab of CAD&CG, Zhejiang University, Hangzhou, China and Hangzhou Research Institute of Holographic and AI Technology, Hangzhou, China; Tianjia Shao, tjshao@zju.edu.cn, State Key Lab of CAD&CG, Zhejiang University, Hangzhou, China and Hangzhou Research Institute of Holographic and AI Technology, Hangzhou, China.



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CCS Concepts: • **Computing methodologies** → **Reconstruction; Rendering; Point-based models.**

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1 Introduction

Significant progress has been achieved in high-fidelity 3D scene reconstruction, particularly following the emergence of Neural Radiance Fields (NeRF) [Mildenhall et al. 2021] and 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023]. While these methods enable the reconstruction of high-quality scenes from dozens of multi-view images, the data acquisition process remains time-consuming and labor-intensive. Furthermore, capturing such datasets is often challenging for novice users who lack expertise in view selection and coverage. Simplifying the capture process is therefore essential to broaden the application of these techniques in fields such as VR/AR, entertainment production, and autonomous robotics.

Some work has focused on 3D reconstruction from sparse-view images to streamline the acquisition process [Li et al. 2024; Wu et al. 2025b,a; Zhong et al. 2025]. These approaches typically initialize low-quality 3D Gaussians from limited input views and subsequently enhance the images rendered from virtual cameras. These repaired images then serve as supervisory signals to refine the 3D Gaussian attributes. By iteratively performing image restoration and Gaussian refinement, these methods aim to produce high-quality scene representations. Early methods rely on different hand-crafted priors to restore Gaussian-rendered images, but are limited to reconstructing a localized region of a scene [Hedman et al. 2021; Li et al. 2024; Niemeyer et al. 2022; Yu et al. 2021].

More recently, using a pretrained diffusion model to iteratively improve the Gaussians has become a popular pipeline, and has achieved the state-of-the-art results [Wu et al. 2025b,a; Zhong et al. 2025]. However, when applying such a pipeline to the task of sparse-view 3D scene reconstruction, we observe that existing methods still struggle to generate high-quality results due to three key challenges that have been overlooked by prior approaches. First, existing methods ignore the view inconsistency problem in diffusion-prior repaired images. While this may be acceptable for small unseen regions, it results in severe blurring and visual artifacts for large unseen regions (see the first row in Fig. 5). Second, previous approaches for scene reconstruction [Wu et al. 2025b,a; Zhong et al. 2025] use pretrained diffusion models without finetuning, leading to unsatisfying results when a large domain gap exists between the diffusion prior and the target scene (see the middle example in Fig. 1). Third, existing frameworks employ simple camera interpolation between the input views [Li et al. 2024; Wu et al. 2025b,a; Zhong et al. 2025], where the generated virtual cameras only cover limited regions around these input viewpoints and fail to provide sufficient coverage of the entire 3D scene, resulting in incomplete reconstructions (see the left example in Fig. 9).

In this paper, we aim to reconstruct high-fidelity 3D scenes from *extremely sparse inputs*, i.e., only 6 to 8 images. To tackle these challenges, we create a specialized diffusion model by finetuning a pretrained diffusion model [Wu et al. 2025b] on the input and corresponding degraded images, adapting the model to the target scene so it can repair scene renderings without domain gap. Then we design a simple but effective camera trajectory planning scheme. For each newly sampled virtual camera, we connect it to its two

nearest neighboring cameras to build a candidate path, and measure the scene coverage of the path. By iteratively retaining only the camera paths that provide sufficient improvement in scene coverage, we obtain a camera trajectory that covers the whole scene. In the key stage of Gaussian refinement, we freeze the diffusion model, measure the view consistency among neighboring repaired images, and inject it as a condition into the sampling process of the frozen diffusion model during Gaussian optimization. It progressively guides the diffusion model to generate more view-consistent images in a training-free manner, finally obtaining high-fidelity 3D Gaussians.

Experiments on various indoor scene datasets demonstrate that our method can reconstruct high-fidelity 3D scenes using only six to eight image captures. Compared to state-of-the-art methods, our approach effectively recovers high-quality scenes without noticeable artifacts.

In summary, the main contributions of our work include:

- A novel framework for reconstructing high-fidelity 3D scenes using only 6-8 image captures.
- A specialized diffusion model for scene image repair without domain gap, a view-consistency conditioned sampling process in the diffusion model for Gaussian refinement, and a camera trajectory planning scheme that generates virtual cameras covering the entire 3D scene.
- Superior performance compared with state-of-the-art methods on various indoor scene datasets in terms of both reconstruction quality and completeness.

2 Related Work

The success of Neural Radiance Fields (NeRF) [Mildenhall et al. 2021] and 3D Gaussian Splatting (3DGS) [Kerbl et al. 2023] has inspired a great amount of follow-up research for 3D scene reconstruction [Barron et al. 2022; Fu et al. 2024; Huang et al. 2024; Kulhanek et al. 2024; Matsuki et al. 2024; Müller et al. 2022; Papantonakis et al. 2024; Peng et al. 2024, 2025; Ren et al. 2024; Roessle et al. 2023; Sabour et al. 2025; Ye et al. 2025; Yu et al. 2024]. [Chen et al. 2025; Gao et al. 2022; Wu et al. 2024] offer comprehensive surveys of recent progress in this area. In this paper, we focus only on works related to sparse-view 3D reconstruction.

2.1 Prior-based 3D scene reconstruction

Early approaches typically leverage priors, such as smoothness losses [Hedman et al. 2021; Yu et al. 2021], sparsity priors [Niemeyer et al. 2022; Yang et al. 2023; Zhang et al. 2024], and depth-normal guidance [Han et al. 2025; Li et al. 2024; Zheng et al. 2025], to repair the Gaussian-rendered images at virtual cameras. For example, Yu et al. [Yu et al. 2021] introduce a smoothness prior on the 3D Gaussian density field to encourage spatially coherent geometry, reducing noise and artifacts during sparse-view optimization. Zhang et al. [Zhang et al. 2024] introduce a sparsity prior on the 3D Gaussian density to suppress unsupported Gaussians and prevent overfitting to sparse views. By encouraging a compact representation, the method reduces spurious geometry in unseen regions and stabilizes the optimization process under sparse-view settings. Li et al. [Li

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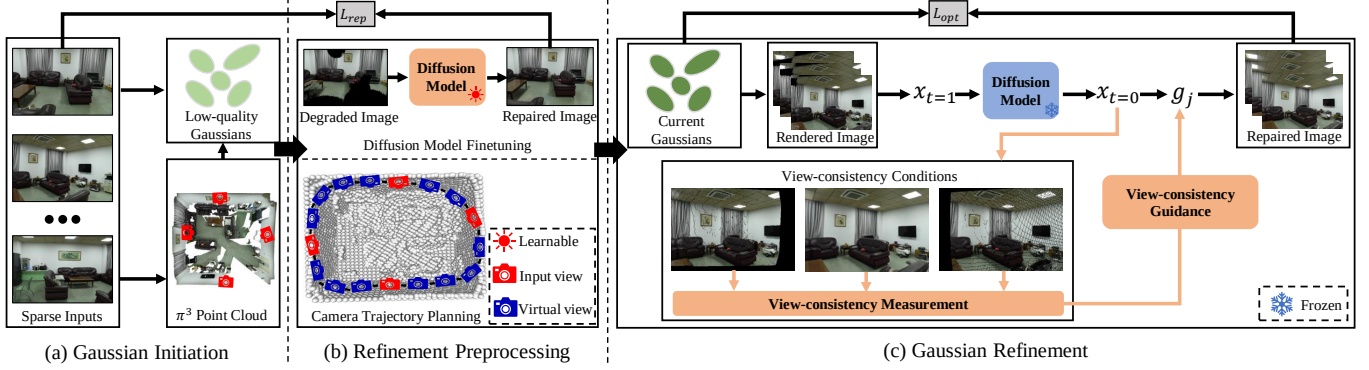


Fig. 2. Overview of our method. (a) We input a sparse set of (e.g., 4) unposed images into a feed-forward visual geometry reconstruction model π^3 [Wang et al. 2025] to estimate camera poses and reconstruct a point cloud, which are then used to obtain an initial set of low-quality 3D Gaussians. (b) We create a specialized diffusion model by finetuning a pretrained diffusion model on the input and corresponding degraded images. Besides, we design a camera trajectory planning scheme to obtain a camera trajectory that covers the whole scene. (c) We repair the Gaussian-rendered images at the planned camera trajectory, and use the repaired images to optimize Gaussians for Gaussian refinement. As the repaired images still have conflicts across different views, which cannot be directly used to generate high-quality Gaussians, we propose a training-free view-consistency conditioned sampling process in the diffusion model for Gaussian refinement.

et al. 2024] leverage global-local depth and normal priors to supervise the optimization of 3D Gaussians in unseen regions, enabling the recovery of accurate scene geometry and detailed appearance from sparse inputs. While promising results are demonstrated, these methods can only deal with small regions.

2.2 Diffusion model based 3D scene reconstruction

Inspired by the powerful generative ability of general diffusion models, recent methods leverage general diffusion models retrained on large-scale real-world scene datasets to repair the Gaussian-rendered images, including image diffusion models [Wu et al. 2025b] and video diffusion models [Bao et al. 2025; Liu et al. 2026; Wu et al. 2025b,a]. However, these methods overlook the fact that the repaired scene images still have conflicts across different views, leading to blurring and apparent artifacts on large unseen regions. Notably, FreeDoM [Yu et al. 2023] is designed to ensure consistency between the input text/image and the output image. VD-3DGS [Zhong et al. 2025] only enforces consistency inside each repaired image, i.e., the consistency between the repaired and observed regions. It ignores the consistency across repaired images, leading to unsatisfying results on large unseen regions. In contrast, we measure the consistency across multi-view images, and use the multi-view consistency to guide the sampling process.

Besides, most methods [Wu et al. 2025b,a; Zhong et al. 2025] use pretrained diffusion models without finetuning, leading to unsatisfying results when large domain gap exists between the diffusion prior and target scene. GaussianObject [Yang et al. 2024] is the only one performing finetuning, but it is designed for object-level reconstruction. Its finetuning strategy (i.e., adding noises on Gaussians) cannot simulate the large unseen regions of scenes.

Beyond sparse-view 3D scene reconstruction methods, several single-image-to-3D approaches use diffusion models to complete Gaussian-rendered images [Ni et al. 2025; Yu et al. 2025; Zhang et al. 2025], but they still suffer from the mentioned three challenges,

which restrict them to reconstructing only small scene regions from a single input view.

3 Method

The overview of our method is illustrated in Fig. 2, which consists of three stages. Given a sparse set of (e.g., 4) unposed images as input, we feed them into the feed-forward visual geometry reconstruction model π^3 [Wang et al. 2025] to estimate camera poses and reconstruct a point cloud P , which are then used to obtain an initial set of low-quality 3D Gaussians G_0 (Sec. 3.2). Next we create a specialized diffusion model by finetuning a pretrained diffusion model [Wu et al. 2025b] on the input and corresponding degraded images, so that the general diffusion model is adapted to the target scene (Sec. 3.3). Besides, we design a camera trajectory planning scheme to obtain a camera trajectory τ_{best} that covers the whole scene (Sec. 3.4). As the repaired images still have conflicts across different views, which cannot be directly used to generate high-quality Gaussians, we propose a training-free view-consistency conditioned sampling process in the diffusion model for Gaussian refinement (Sec. 3.5). We next provide a brief overview of 3DGS and diffusion models (Sec. 3.1), after which we describe each component of our framework in detail.

3.1 Preliminary

3.1.1 3D Gaussian Splatting. 3DGS [Kerbl et al. 2023] explicitly models a 3D scene with a collection of anisotropic Gaussians. Each primitive is parameterized by a tuple $\{\mu, \mathbf{s}, \mathbf{q}, \alpha, \mathbf{c}\}$, comprising the position $\mu \in \mathbb{R}^3$, scaling factors $\mathbf{s} \in \mathbb{R}^3$, rotation quaternion $\mathbf{q} \in \mathbb{R}^4$, opacity $\alpha \in \mathbb{R}$, and view-dependent color coefficients $\mathbf{c}$. The spatial influence of the i -th Gaussian is defined as:

$$G_0(\mathbf{v}) = \exp\left(-\frac{1}{2}(\mathbf{v} - \mu_i)^\top \Sigma_i^{-1}(\mathbf{v} - \mu_i)\right) \quad (1)$$

where the covariance matrix Σ_i is derived from the scaling and rotation parameters $\mathbf{s}_i$ and $\mathbf{q}_i$. For rendering, 3D Gaussians are projected onto the image plane to form 2D splats, denoted as G'_0 ,

which are then composited via front-to-back α -blending. The final pixel color $C(\mathbf{v}_p)$ is formulated as:

$$C(\mathbf{v}_p) = \sum_{i \in \mathcal{K}} c_i \sigma_i \prod_{j=1}^{i-1} (1 - \sigma_j), \quad \sigma_i = \alpha_i G'_0(\mathbf{v}_p), \quad (2)$$

where $\mathcal{K}$ denotes the ordered set of Gaussians overlapping pixel $\mathbf{v}_p$, sorted by depth.

3.1.2 Diffusion model. Diffusion models first corrupt clean samples by progressively adding Gaussian noise and subsequently adopt a denoising function to reconstruct the original data by the sampling process [Blattmann et al. 2023; Ho et al. 2020, 2022; Rombach et al. 2022; Saharia et al. 2022; Zhang et al. 2023]. The underlying principle of standard diffusion models involves employing a noise estimator ϵ_θ , typically a U-Net [Ronneberger et al. 2015], to predict the noise injected during the iterative T -step sampling process [Ho et al. 2020; Yu et al. 2023] as follows:

$$\mathbf{x}_{t-1} = (1 + \beta_t/2) \mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \sqrt{\beta_t} \mathbf{z}, \quad (3)$$

where $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ represents the estimated score function derived from $\epsilon_\theta(\mathbf{x}_t, t)$. In this formulation, β_t denotes predefined variance parameters and $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ accounts for the stochastic noise component. Given condition Q , SDE [Song et al. 2021] modifies the original score function $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ in Eq. 3 to a conditional score function $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | Q)$ to guide the sampling process, which can be expanded using the Bayesian rule as:

$$\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | Q) = \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p(Q | \mathbf{x}_t), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log p(Q | \mathbf{x}_t)$ acts as a correction gradient so that the generated results are compatible with the given condition Q . If Q is defined as the view-consistency condition, $\nabla_{\mathbf{x}_t} \log p(Q | \mathbf{x}_t)$ can be used to guide the diffusion model sampling toward generating view-consistent images.

3.2 Gaussian Initiation

Given unposed inputs, the initial low-quality Gaussians G_0 typically rely on point clouds and camera poses estimated by Structure-from-Motion (SfM) [Garcia-Fidalgo and Ortiz 2017; Schönberger et al. 2016; Schönberger and Frahm 2016]. However, traditional SfM yields inaccurate camera pose estimates and incomplete or noisy point cloud for sparse inputs. Alternatively, we adopt the π^3 model [Wang et al. 2025], a recent feed-forward visual geometry reconstruction framework for estimating affine-invariant camera poses and scale-invariant point clouds from sparse inputs, as illustrated in Fig. 2 (a). π^3 gives a higher-quality initial point cloud and more reliable camera poses. Based on its prediction, we obtain the initial Gaussians G_0 by minimizing:

$$\mathcal{L}_{opt} = \|\mathbf{x}_{gt} - \mathbf{x}_r(G_0)\|_1 + (1 - SSIM(\mathbf{x}_{gt}, \mathbf{x}_r(G_0))) \quad (5)$$

where $\mathbf{x}_{gt}$ and $\mathbf{x}_r$ denote the sparse input images and the rendered images, respectively. G_0 is then used to finetune the general diffusion model and serves as the initialization for the Gaussian refinement. Meanwhile, the generated π^3 point cloud is utilized for subsequent camera trajectory planning.

ALGORITHM 1: Camera Trajectory Planning.

```

1: input: incomplete poisson mesh, input cameras  $\tau_{input} = \{\tau_j\}_{j=1}^m$ ,
   added threshold  $\varphi$ , maximum step  $Q_{end}$ , interpolation threshold  $d_\varphi$ 
2: output: best trajectory  $\tau_{best}$ 
3: Sample spheres  $\{S_i\}_{i=1}^N$  on the poisson mesh and its OBB.
4: Sample points  $\{s_i^k\}_{k=1}^{N'}$  on each sphere  $S_i$ .
5: while  $Q < Q_{end}$  do
6:    $\tau_k \leftarrow$  Randomly sample a camera in OBB
7:    $\{\tau_{k_1}, \tau_{k_2}\} \leftarrow$  Find two nearest cameras in  $\tau_{input}$  via Eq.9
8:    $m_1 \leftarrow \text{Int}(\text{Dist}(\tau_{k_1}, \tau_k)/d_\varphi)$ ,  $m_2 \leftarrow \text{Int}(\text{Dist}(\tau_{k_2}, \tau_k)/d_\varphi)$ 
9:    $\{\tau_{k_1=k-m_1}, \dots, \tau_k, \dots, \tau_{k_2=k+m_2}\} \leftarrow$  Trajectory by Eq. 10
10:   $g_w \leftarrow \text{Gain}(\tau_{k_1}, \dots, \tau_k, \dots, \tau_{k_2} | \tau_{input})$ 
11:  if  $g_w > \varphi$  then
12:     $\tau_{input} \leftarrow \tau_{input} \cup \{\tau_{k_1}, \dots, \tau_k, \dots, \tau_{k_2}\}$ 
13:     $Q \leftarrow 0$ 
14:  else
15:     $Q \leftarrow Q + 1$ 
16:  end if
17: end while
18:  $\tau_{best} \leftarrow$  Output a camera trajectory  $\tau_{input}$  with sufficient coverage

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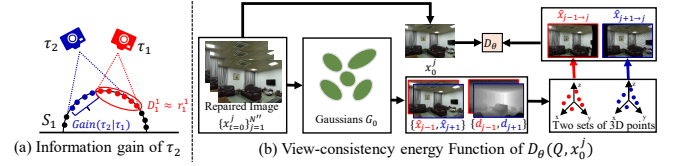


Fig. 3. Detailed illustrations of the computation of the information gain (Sec. 3.4) and the view-consistency energy function (Sec. 3.5).

3.3 Diffusion Model Finetuning

Although the input views are sparse, they capture informative appearance characteristics of the scene. We therefore leverage these images to finetune a specialized diffusion model tailored to the scene context. Specifically, we finetune the model using training pairs generated by degrading the available input views. We propose a masking technique for rendered images to represent occluded or invisible areas. This approach effectively creates a mapping between the degraded observations and the original scene content for scene-specific adaptation. As in [Yang et al. 2024], we add random noise to the attributes of G_0 and render noise-degraded images from the noisy Gaussians at all reference views. Those noisy Gaussians alone cannot model occluded and invisible regions. To this end, we introduce masks by randomly removing a subset of Gaussians to simulate degraded images with large occlusion for better finetuning. Following [Sauer et al. 2024; Wu et al. 2025b], we finetune the model with a frozen VAE encoder and a LoRA-adapted decoder. The overall loss function used to supervise the training process is defined as:

$$\mathcal{L}_{rep} = \|R(\mathbf{x}_{deg}) - \mathbf{x}_{gt}\|_2 + \mathcal{L}_{gram} + \mathcal{L}_{LPIPS}, \quad (6)$$

where we adopt an L_2 loss and a perceptual LPIPS loss between the model prediction $R(\mathbf{x}_{deg})$ and the sparse input $\mathbf{x}_{gt}$, along with a Gram matrix loss $\mathcal{L}_{gram}$ [Wu et al. 2025b].

3.4 Camera Trajectory Planning

To ensure sufficient coverage of the full 3D scene, inspired by [Roberts and Hanrahan 2016; Xie et al. 2018; Yang et al. 2018; Zhang et al. 2021], we propose a camera trajectory planning scheme. We first define an information gain metric to quantify the contribution of each camera to scene coverage, followed by a description of how this metric is used for camera trajectory planning.

3.4.1 Information gain. We first adopt Poisson surface reconstruction [Kazhdan et al. 2006] to convert the π^3 point cloud P into a surface mesh. As the π^3 point cloud P is generated from sparse views, the reconstructed surface mesh may not be watertight. If we directly follow [Kazhdan et al. 2006; Roberts et al. 2017] and sample points only from the mesh surface, these incomplete regions cannot be represented. To solve it, we sample more points from the six faces of an oriented bounding box (OBB) that tightly encloses the coarse mesh to cover missing regions. We convert all of the N surface points to spheres $\{S_i\}_{i=1}^N$ to cover the scene (see Fig. 2 (b)). After that, we compute the intersection area between these spheres $\{S_i\}_{i=1}^N$ and the view frustum of each camera τ_j to evaluate how well τ_j contributes to the overall scene coverage. Concretely, the coverage of the local area around S_i from camera τ_j is denoted as D_i^j , which corresponds to the intersection between the viewing frustum of τ_j and the sphere S_i . The effective area of D_i^j depends on both the distance between τ_j and S_i , as well as the elevation angle of the camera relative to the surface. To simplify the computation of the intersection area, we uniformly sample N' surface points $\{s_i^k\}_{k=1}^{N'}$ on each sphere S_i , where each point s_i^k covers a local sphere area. As shown in Fig. 3 (a), the coverage of the local area by D_i^j can be represented as the collection of sample points r_i^j on the sphere S_i that fall within the view frustum of τ_j :

$$D_i^j \approx r_i^j = \{s_i^k \mid s_i^k \in S_i \wedge s_i^k \in D_i^j\}_{k=1}^{N'} \quad (7)$$

if τ_j is defined as a candidate camera to be appended to the planned camera path, which consists of a sequence of cameras $\{\tau_1, \dots, \tau_{j-1}\}$. The contribution of τ_j to covering all 3D contents, also called *information gain* [Kazhdan et al. 2006], is defined as the amount of added scene information. It denotes the number of new sample points on each local area D_i^j that can be observed by τ_j beyond the observations from the previous cameras $\{\tau_1, \dots, \tau_{j-1}\}$:

$$\text{Gain}(\tau_j | \tau_{j-1}, \dots, \tau_1) = \sum_i \left| r_i^j \setminus \bigcup_{l=1}^{j-1} r_i^l \right| \quad (8)$$

3.4.2 Camera trajectory planning. The detailed camera trajectory planning scheme is outlined in Alg. 1. We first record the information gain of the input cameras $\tau_{input} = \{\tau_j\}_{j=1}^m$, and randomly sample a new camera τ_k in OBB. Then, we compute the pose distance between the new camera τ_k and each camera in the set τ_{input} , which can be formulated as:

$$\text{Dist}(\tau_1, \tau_2) = w_t \cdot \|t_1 - t_2\|_2 + w_r \cdot \arccos(2(|q_1 \cdot q_2|)^2 - 1), \quad (9)$$

where τ_1 and τ_2 denote the poses of two cameras. w_t and w_r represent the weights for the translation and rotation distances, respectively. t_1 and t_2 denote the translation vectors, while q_1 and q_2 represent the rotation vectors. According to Eq. 9, we obtain the

ALGORITHM 2: View-consistency Conditioned Sampling Process.

```

1: input: initial low-quality Gaussians  $G_0$ , trajectory  $\tau_{best}$ , refine steps  $r_t$ 
2: output: output 3DGS  $G^*$ 
3: for  $i = r_t, \dots, 1$  do
4:    $\{x_{t=1}^j\}_{j=1}^{N''} \leftarrow \text{Render } N'' \text{ images by } G_0 \text{ via } \tau_{best}$ 
5:   for  $j = N'', \dots, 1$  do
6:      $x_0^j \leftarrow (1 + \beta_1/2) x_1^j + \beta_1 \nabla_{x_1} \log p(x_1^j) + \sqrt{\beta_1} z$ 
7:   end for
8:    $G_0 \leftarrow \text{Optimize 3DGS using } \{x_0^j\}_{j=1}^{N''}$ 
9:   if  $i=1$  then
10:    break
11:   end if
12:   for  $j = N'', \dots, 1$  do
13:      $g_j \leftarrow \nabla_{x_{t=1}^j} \mathcal{D}_\theta(Q, x_{t=0}^j)$ 
14:      $x_0^j \leftarrow x_0^j - \rho_j g_j$ , where  $\rho_j$  is a predefined scale factor
15:   end for
16:    $G_0 \leftarrow \text{Optimize 3DGS using } \{x_0^j\}_{j=1}^{N''}$ 
17: end for
18:  $G^* \leftarrow \text{Output 3DGS } G_0$ 

```

two cameras $\{\tau_{k_1}, \tau_{k_2}\}$ in τ_{input} with the smallest pose distances to τ_k , and interpolate a trajectory of length $m_1 + m_2 + 1$ between τ_k and $\{\tau_{k_1}, \tau_{k_2}\}$ as follows:

$$\{\tau_{k_1}, \dots, \tau_k, \dots, \tau_{k_2}\} = \text{inter}(\tau_{k_1}, \tau_k) \cup \text{inter}(\tau_k, \tau_{k_2}). \quad (10)$$

We extend Eq. 8 to estimate the information gain g_w of the interpolated trajectory $\{\tau_{k_1}, \dots, \tau_k, \dots, \tau_{k_2}\}$ and use it to determine whether the entire interpolated trajectory should be included in the final trajectory. If the information gain g_w exceeds a predefined threshold φ , all cameras in the interpolated trajectory are considered to provide sufficient new information and are therefore incorporated into the set τ_{input} . This process is iteratively repeated until no additional cameras can be found that contribute new information to the scene. Thus, τ_1 and τ_2 in the updated τ_{input} are not only input poses, but are also already sampled and adopted virtual camera poses. The interpolation among these cameras randomly sampled inside the scene OBB can cover more regions than existing methods. Finally, we obtain the optimal trajectory τ_{best} , which allows a small number of cameras to fully reconstruct the 3D scene, while ensuring that each planned camera receives complementary guidance information from two neighboring repaired images, thereby propagating view consistency across all planned cameras (see Sec. 3.5).

3.5 Gaussian Refinement

Previous work, e.g., Difix3D+ and GenFusion, directly conditions the diffusion model on multiple views. However, as mentioned above, these methods cannot guarantee the view consistency cross multi-view images. Hence, we propose a training-free view-consistency conditioned sampling process in the diffusion model for Gaussian refinement. During Gaussian refinement, we freeze the diffusion model, measure the view consistency among neighboring repaired images, and inject it as a condition into the sampling process of the frozen diffusion model. Doing so progressively guides the diffusion model to generate more view-consistent images, finally obtaining high-fidelity 3D Gaussians G^* .

3.5.1 Diffusion guidance. As discussed in Sec. 3.1, by defining the view-consistency condition as Q , the corresponding conditional score term $\nabla_{\mathbf{x}_t} \log p(Q | \mathbf{x}_t)$ in Eq. 4 can be used to enforce view consistency among the repaired images. Inspired by [Yu et al. 2023], we adopt a training-free manner to inject the view-consistency condition into the sampling process in the diffusion model. Especially, the term $\nabla_{\mathbf{x}_t} \log p(Q | \mathbf{x}_t)$ can further be modeled as a consistency-guided energy function [LeCun et al. 2006; Yu et al. 2023; Zhao et al. 2022]:

$$p(Q | \mathbf{x}_t) = \frac{\exp\{-\lambda \mathcal{E}(Q, \mathbf{x}_t)\}}{Z}, \quad (11)$$

where λ denotes a positive temperature coefficient and $Z > 0$ is a normalization constant. $\mathcal{E}(Q, \mathbf{x}_t)$ represents an energy function that quantifies how well the current sample $\mathbf{x}_t$ aligns with the target of view-consistency. The energy function between Q and $\mathbf{x}_t$ can be further approximated by the time-dependent energy function between Q and $\mathbf{x}_0$ [Yu et al. 2023] as follows:

$$\mathcal{E}(Q, \mathbf{x}_t) \approx \mathcal{D}_\theta(Q, \mathbf{x}_0 | t). \quad (12)$$

With Eq. 11 and Eq. 12, it can be written as:

$$\nabla_{\mathbf{x}_t} \log p(Q | \mathbf{x}_t) \propto -\nabla_{\mathbf{x}_t} \mathcal{D}_\theta(Q, \mathbf{x}_0). \quad (13)$$

Notably, we adopt the single-step (i.e., $T = 1$) diffusion model Difix3D+ [Wu et al. 2025b], which takes the images rendered from the initial low-quality Gaussians G_0 as inputs, denoted as $\{\mathbf{x}_{t=1}^j\}_{j=1}^{N''} = \{\mathbf{x}_{t=1}^1, \dots, \mathbf{x}_{t=1}^{N''}\}$, to generate the repaired images $\{\mathbf{x}_{t=0}^j\}_{j=1}^{N''}$. Thus, the problem can be reformulated as defining a suitable energy function $\mathcal{D}_\theta(Q, \mathbf{x}_0)$ that effectively measures the consistency among the multi-view repaired images $\{\mathbf{x}_0^j\}_{j=1}^{N''}$.

3.5.2 Energy formulation. In order to build such a consistency measurement function between $\{\mathbf{x}_0^j\}_{j=1}^{N''}$, we establish pixel-wise correspondences across views and then measure the discrepancies between corresponding pixels. Reliable correspondences can be derived from depth maps and associated camera poses [Mildenhall et al. 2021; Schönberger and Frahm 2016; Toft et al. 2020]. We leverage these geometric cues to project pixels into 3D points and unproject them across views to construct accurate cross-view correspondences. As shown in Fig. 3 (b), we first use $\{\mathbf{x}_0^j\}_{j=1}^{N''}$ to optimize 3DGS G_0 and then render corresponding RGB images $\{\hat{\mathbf{x}}_j\}_{j=1}^{N''}$ and depth images $\{d_j\}_{j=1}^{N''}$. For each image $\mathbf{x}_0^j$, we unproject its two neighboring rendered images $\hat{\mathbf{x}}_{j-1}$ and $\hat{\mathbf{x}}_{j+1}$ using their corresponding depth maps d_{j-1} and d_{j+1} , together with the camera poses, to obtain two sets of 3D RGB points. These points are then projected onto the image plane of $\mathbf{x}_0^j$ to generate two point-rendered images $\hat{\mathbf{x}}_{j-1 \rightarrow j}$ and $\hat{\mathbf{x}}_{j+1 \rightarrow j}$. $\hat{\mathbf{x}}_{j-1 \rightarrow j}$ and $\hat{\mathbf{x}}_{j+1 \rightarrow j}$ provide complementary multi-view information for $\mathbf{x}_0^j$. Therefore, we typically select these two neighboring point-rendered images for view-consistency measurement. The final consistency energy function is formulated as the pixel-wise L1 loss between each original repaired image $\mathbf{x}_0^j$ and its corresponding neighboring point-rendered images $\{\hat{\mathbf{x}}_{j-1 \rightarrow j}, \hat{\mathbf{x}}_{j+1 \rightarrow j}\}$:

$$\mathcal{D}_\theta(Q, \mathbf{x}_0^j) = \left\| \mathbf{M}_{j-1} \odot (\mathbf{x}_0^j - \hat{\mathbf{x}}_{j-1 \rightarrow j}) \right\|_1 + \left\| \mathbf{M}_{j+1} \odot (\mathbf{x}_0^j - \hat{\mathbf{x}}_{j+1 \rightarrow j}) \right\|_1, \quad (14)$$

where $\mathbf{M}_{j-1}$ and $\mathbf{M}_{j+1}$ denote the validity masks corresponding to the point-rendered images $\hat{\mathbf{x}}_{j-1 \rightarrow j}$ and $\hat{\mathbf{x}}_{j+1 \rightarrow j}$, respectively, and $\odot$

represents element-wise multiplication. As shown in Alg. 2, this view-consistency energy function can inject view-consistency conditions into the sampling process during Gaussian optimization, to progressively produce more view-consistent images (See Fig. 7) in a training-free manner.

4 Experiments

4.1 Experimental Setup

4.1.1 Baselines and metrics. We compare S2C-3D with recent sparse-view 3D scene reconstruction methods, including DNGaussian [Li et al. 2024], Difix3D+ [Wu et al. 2025b], GenFusion [Wu et al. 2025a], and VD-3DGS [Zhong et al. 2025]. For quantitative evaluation of novel view synthesis, we report PSNR, SSIM [Wang et al. 2004], and LPIPS [Zhang et al. 2018].

4.1.2 Implementation details. For diffusion model finetuning, we adopt Difix3D+ [Wu et al. 2025b] as the baseline and set the number of training steps to 3000. We use the AdamW optimizer with a constant learning rate of 2×10^{-5} and a linear warmup of 500 steps. All stages are trained on a RTX 4090 GPU with a 9950X3D CPU. The parameters are set as follows: the added threshold $\varphi = 0.1$, the maximum step $Q_{\text{end}} = 30$, the interpolation threshold $d_\varphi = 0.2$, the refinement steps $r_t = 4$, the consistency weight $\rho_j = 1$ and the distance weights $w_t = 1$ and $w_r = 1$.

4.1.3 Datasets. We conduct experiments on all 8 scenes from the synthetic Replica dataset [Straub et al. 2019], 100 scenes from the real-world ScanNet++ dataset [Yeshwanth et al. 2023], and 11 real-world scenes captured by ourselves, termed S2C-Scene. The training-view settings of ScanNet++ are set to 6, 7 and 8 views. And the settings of Replica and S2C-Scene are set to 4, 5 and 6 views.

4.2 Evaluation of Scene Reconstruction

We qualitatively compare the rendering quality for novel view synthesis across all compared methods, as shown in Tab. 1. We also evaluate the scene completeness on the ScanNet++ dataset in the supplementary materials. The results show that our approach consistently achieves the best performance across all evaluation metrics, indicating superior reconstruction quality and scene completeness.

Figs. 4 and 5 show novel view synthesis results on the ScanNet++ and S2C-Scene dataset across different views. We observe that all compared methods struggle to reconstruct high-fidelity scenes from highly sparse images. Especially, the results of Difix3D+, GenFusion, and VD-3DGS still contain large incomplete regions (e.g., the ceiling in the fourth row of Fig. 4). This is because the large domain gap between the training data and the target scene leads to a substantial repair performance drop (see examples of incomplete repaired images produced by Difix3D+ in Fig. 8). Besides, their simple interpolation between the input views cannot generate virtual cameras to cover these occluded and invisible regions (see Fig. 9). By leveraging a specialized diffusion model for scene image repair without domain gap and a camera trajectory planning scheme that generates virtual cameras covering the entire 3D scene, our method enables complete 3D scene reconstruction. Moreover, Genfusion tends to generate blurred Gaussians (e.g., the chair in the third row of Fig. 4), and VD-3DGS often produces low-quality Gaussians with extensive

Table 1. Quantitative comparisons of sparse-view 3D reconstruction methods on the ScanNet++, Replica, and S2C-Scene datasets. Our approach demonstrates strong performance across diverse domains and outperforms all compared methods. We highlight the results as follows: **best**, **second best**, and **third best**.

Dataset	Method	6-view			7-view			8-view		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ScanNet++	3DGS [Kerbl et al. 2023]	13.063	0.496	0.465	13.911	0.517	0.452	14.956	0.554	0.432
	DNGaussian [Li et al. 2024]	12.834	0.420	0.571	13.338	0.436	0.560	13.728	0.446	0.551
	Difix3D+ [Wu et al. 2025b]	13.321	0.549	0.405	14.228	0.575	0.386	15.300	0.613	0.361
	Genfusion [Wu et al. 2025a]	16.605	0.626	0.371	17.139	0.661	0.355	17.993	0.683	0.331
	VD-3DGS [Zhong et al. 2025]	16.536	0.644	0.395	17.488	0.667	0.368	18.291	0.688	0.346
	Ours	17.779	0.683	0.344	17.990	0.690	0.333	18.777	0.706	0.315
Dataset	Method	4-view			5-view			6-view		
		PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Replica	3DGS [Kerbl et al. 2023]	13.577	0.487	0.416	14.673	0.545	0.379	16.117	0.576	0.370
	DNGaussian [Li et al. 2024]	14.264	0.460	0.517	15.010	0.476	0.517	15.945	0.511	0.504
	Difix3D+ [Wu et al. 2025b]	13.498	0.521	0.371	14.745	0.576	0.329	16.297	0.617	0.312
	Genfusion [Wu et al. 2025a]	17.148	0.641	0.353	18.764	0.685	0.305	19.855	0.701	0.290
	VD-3DGS [Zhong et al. 2025]	17.017	0.618	0.387	18.437	0.675	0.340	19.592	0.699	0.314
	Ours	18.445	0.685	0.329	19.917	0.714	0.290	20.360	0.721	0.281
S2C-Scene	3DGS [Kerbl et al. 2023]	13.131	0.448	0.511	14.015	0.473	0.501	14.264	0.482	0.496
	DNGaussian [Li et al. 2024]	13.033	0.428	0.586	13.314	0.442	0.577	13.469	0.444	0.572
	Difix3D+ [Wu et al. 2025b]	13.302	0.493	0.460	14.326	0.533	0.438	14.571	0.544	0.429
	Genfusion [Wu et al. 2025a]	16.029	0.614	0.410	16.774	0.642	0.391	17.188	0.648	0.382
	VD-3DGS [Zhong et al. 2025]	15.902	0.613	0.429	15.932	0.614	0.416	16.341	0.623	0.410
	Ours	16.474	0.634	0.401	17.226	0.648	0.380	17.647	0.658	0.368

Table 2. Ablation study. w/o and w denote without and with models, respectively. *Fine.*, *Traj.*, and *Cond.* denote the proposed diffusion model finetuning, the camera trajectory planning scheme, and the view-consistency conditioned sampling process, respectively.

Model	<i>Fine.</i>	<i>Traj.</i>	<i>Cond.</i>	PSNR $\uparrow$	Replica SSIM $\uparrow$	LPIPS $\downarrow$
Full Model	✓	✓	✓	20.360	0.721	0.281
w/o Noise Finetuning	✓	✓	✓	19.815	0.714	0.295
w/o Mask Finetuning	✓	✓	✓	18.522	0.653	0.336
w Difix3D+	×	✓	✓	18.160	0.640	0.349
w/o Camera Trajectory Planning	✓	×	✓	18.253	0.645	0.342
w/o View-consistency Conditions	✓	✓	×	18.129	0.639	0.353

artifacts (e.g., the table and background in the third row of Fig. 5). Both issues can also be observed in the 3D Gaussians generated by Difix3D+, such as the table in the third row of Fig. 5. This is because they overlook the fact that the repaired scene images still have conflicts across different views. While our method proposes a view-consistency conditioned sampling process to progressively guide the diffusion model to generate more view-consistent images, finally obtaining high-fidelity 3D Gaussians.

4.3 Ablation Study

In this section, we isolate our algorithmic choices and evaluate their effects on the 6-view setting of the Replica dataset.

4.3.1 The effect of diffusion model finetuning. To validate the effect of the specialized diffusion model, we conduct experiments on three models, e.g., without noise training pairs for finetuning, without mask training pairs for finetuning, and directly using the original Difix3D+ model without finetuning, as shown in the third to fifth

rows of Tab. 2. It can be seen that the proposed specialized diffusion model achieves the best performance. As shown in Fig. 8, removing noise training pairs leads to degraded repair quality, while removing mask training pairs, or directly using Difix3D+, fails to repair large visible and occluded regions in the Gaussian-rendered images. In conclusion, the proposed specialized diffusion model can adapt the model to the target scene so it can repair Gaussian renderings without domain gap to obtain high-quality Gaussians (see the first row of Fig. 6).

4.3.2 The effect of camera trajectory planning. As shown in the sixth row of Tab. 2, removing the proposed camera trajectory planning scheme and using simple interpolation between input views to generate virtual cameras for Gaussian refinement leads to a significant performance drop. As shown in Fig. 9, the virtual cameras generated by our camera trajectory planning scheme cover more 3D content than those obtained by simple camera interpolation, resulting in more complete scene reconstruction (see the second and third rows of Fig. 6).

4.3.3 The effect of view-consistency conditioned sampling process. As shown in the seventh row of Tab. 2, removing the proposed view-consistency conditioned sampling process and directly using the repaired images from the finetuned diffusion model for Gaussian refinement results in a significant performance drop. As shown in Fig. 7, the original repaired images still have conflicts across different views, which cannot be directly used to generate high-quality Gaussians. While the proposed training-free view-consistency conditioned sampling process progressively guides the diffusion model

to generate more view-consistent images (see Fig. 7), finally obtaining high-quality Gaussians without blurring or extensive artifacts (see the second and third rows of Fig. 6).

5 Discussion and Conclusion

Our method focuses on indoor scenes with closed areas. For outdoor scenes with open areas, similar to existing methods, our method only reconstructs the regions among the input views. Besides, the sampled cameras can occasionally fall into undesirable positions, e.g., inside objects, and under tables looking upward at the bottom surfaces. The corresponding regions have lower quality due to the limited information from the given views. But since these areas are less important and don't affect the main areas, it is acceptable to leave the undesired cameras there. Our method will fail on concave rooms, producing unexpected regions outside the groundtruth walls. The proposed camera trajectory planning scheme is a greedy algorithm. We will explore a better solution in future.

In conclusion, We propose a novel sparse-view 3D reconstruction framework, termed S2C-3D, to enable high-fidelity and complete 3D scene reconstruction from as few as 6-8 image captures. Our framework features a specialized diffusion model for scene image repair without domain gap, a training-free view-consistency conditioned sampling process in the diffusion model for Gaussian refinement, and a camera trajectory planning scheme that generates virtual cameras covering the entire 3D scene.

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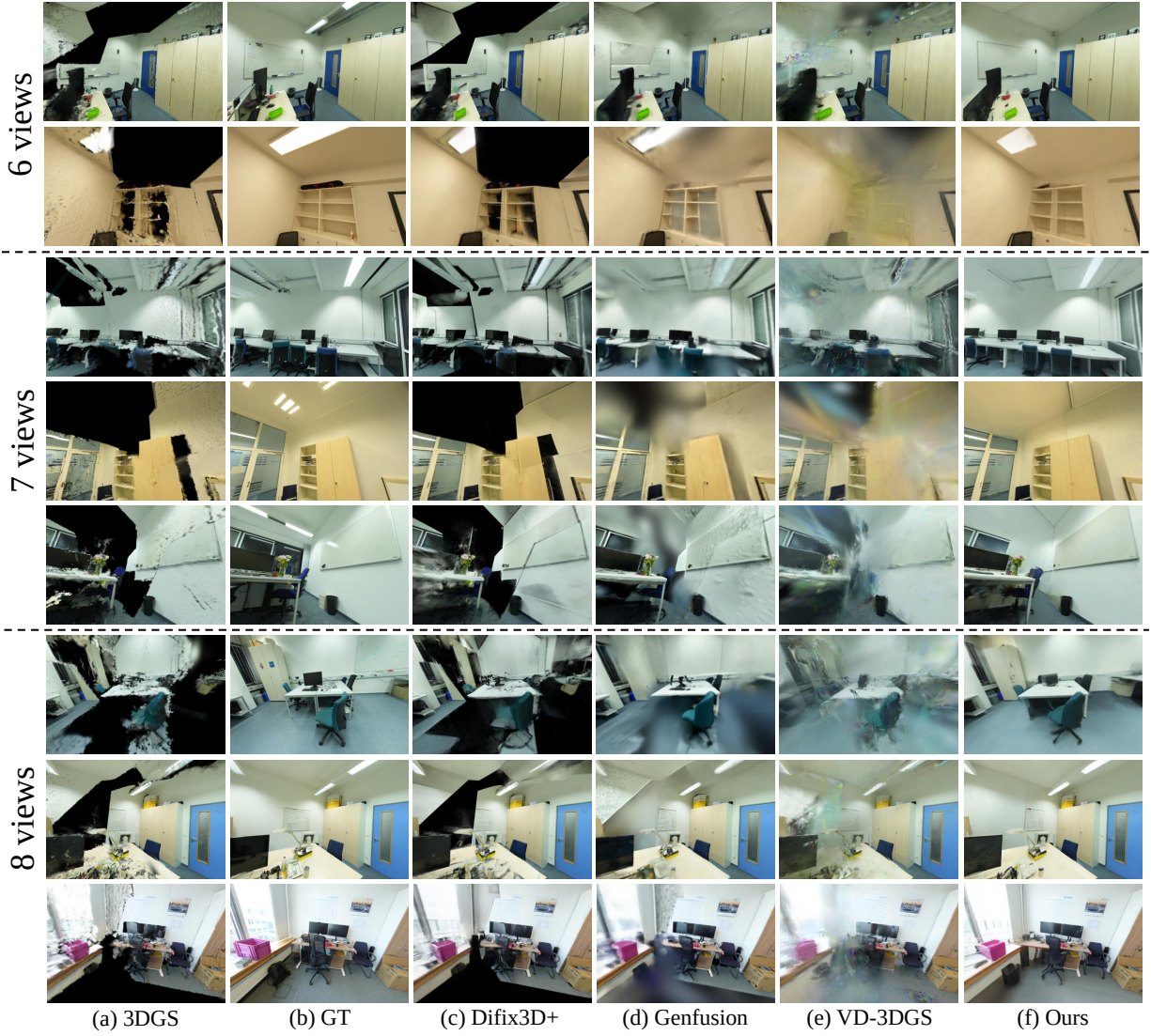


Fig. 4. Qualitative comparisons of novel view synthesis on the ScanNet++ dataset.

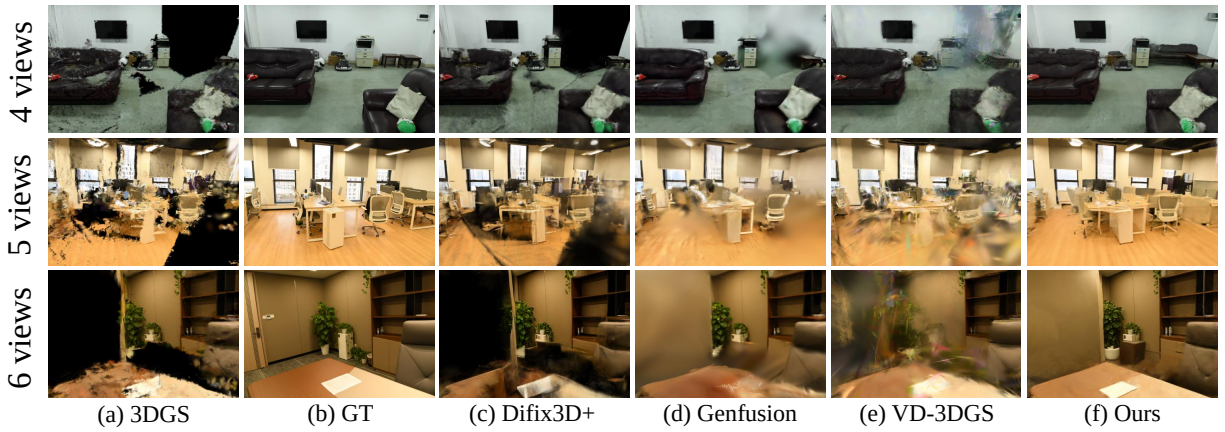


Fig. 5. Qualitative comparisons of novel view synthesis on the S2C-Scene dataset.

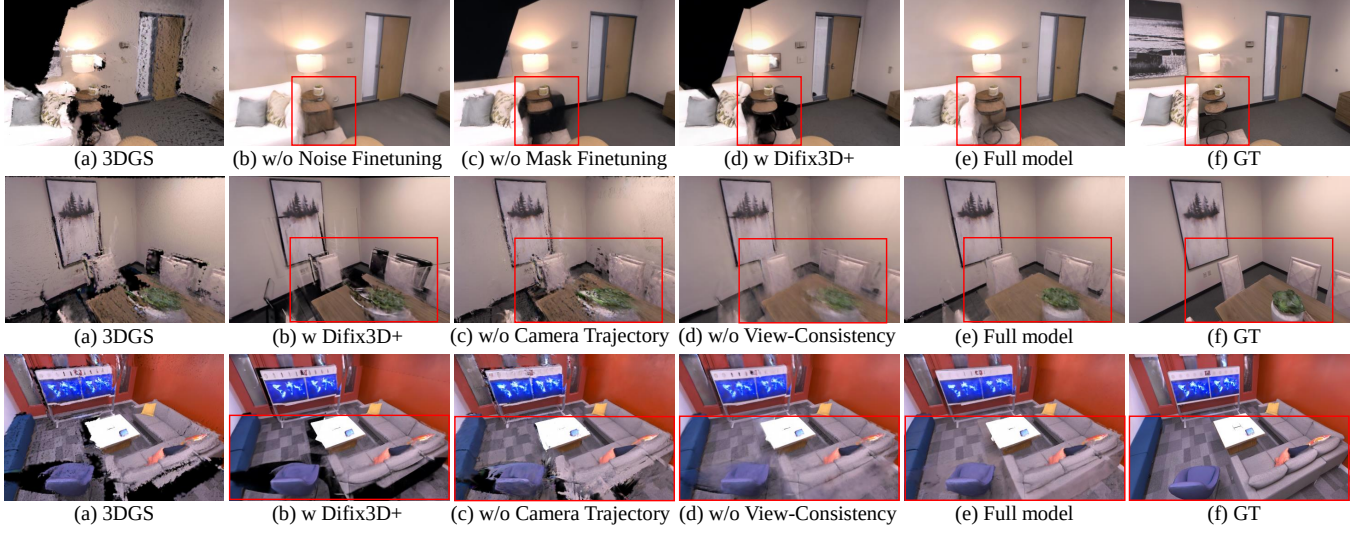


Fig. 6. Ablation study of reconstructed scenes by different models.

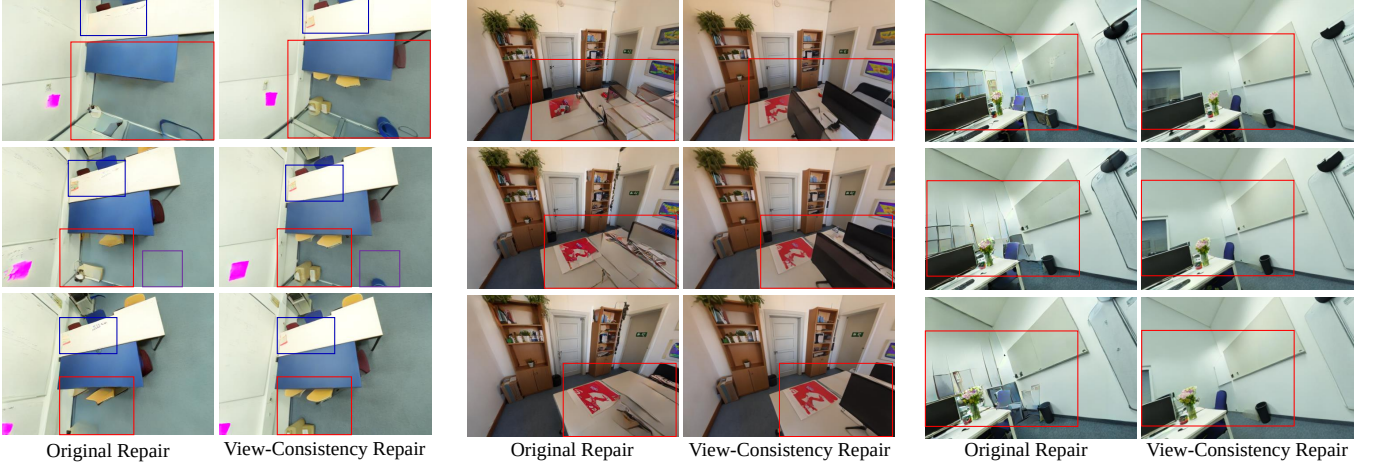


Fig. 7. Qualitative comparison between images repaired by the finetuned diffusion model and those produced by the view-consistency conditioned diffusion process.

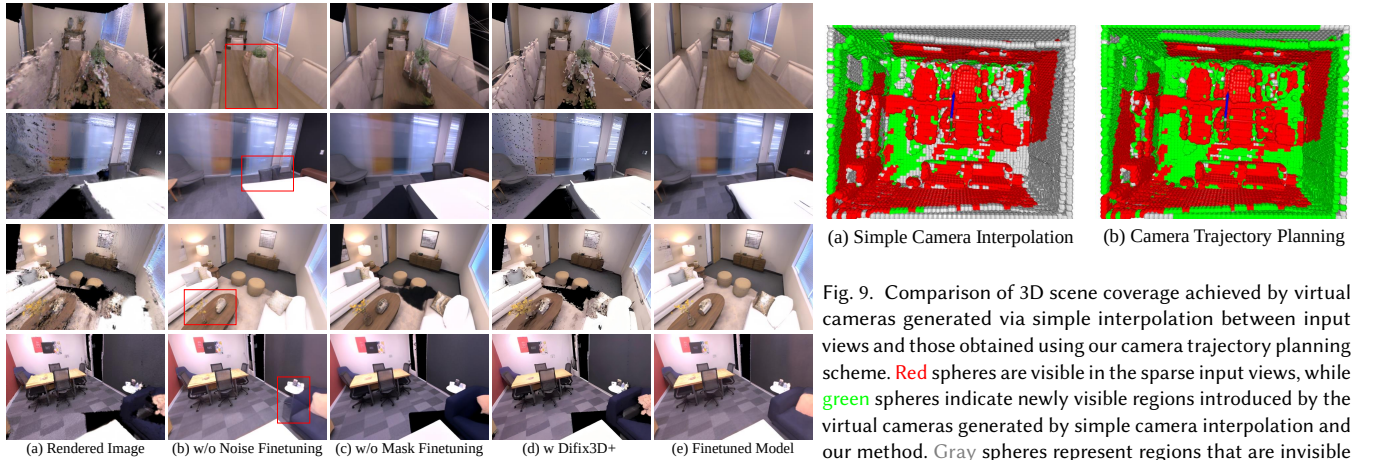


Fig. 8. Qualitative comparison between images repaired by different diffusion models.

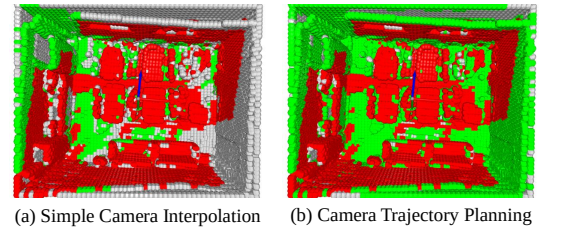


Fig. 9. Comparison of 3D scene coverage achieved by virtual cameras generated via simple interpolation between input views and those obtained using our camera trajectory planning scheme. Red spheres are visible in the sparse input views, while green spheres indicate newly visible regions introduced by the virtual cameras generated by simple camera interpolation and our method. Gray spheres represent regions that are invisible to all cameras.